Dear *eLife* Editors,

We submit our manuscript, *Living Alignment*, for consideration as a Review Article at *eLife*.

How the review moves the field forward

Our paper first highlights an observation that has been made independently in several disciplines: living systems are healthy and flourishing when they avoid collapse to patterns of organization that admit excessively simple descriptions. We work through examples to examine what it means for life to avoid collapse at a range of scales—evolutionary biology, cell biology, neuroscience, psychology and cognitive science, sociology, ecology. We are, as far as we know, the first to review this common pattern across fields.

The paper then applies this insight from the life sciences to the problem of AI alignment. Although humans disagree on details of how aligned AI systems should behave, there is broad agreement that the goal of alignment is futures full of health and flourishing. We argue that the science of life gives direction to what an aligned long-term future looks like even under radical changes in the organization of life on Earth—and how to achieve it. This application of insights from the life sciences to the problem of AI alignment is also an advance.

How it differs from recent examples of reviews in the field

To our knowledge, no other reviews have attempted this kind of synthesis. Our paper builds on broad theories of living systems, such as Kirchhoff et al. (2018); Luisi (2003); Moreno and Mossio (2015), which describe mechanistically how organisms regulate their states and produce new structures. We also build on domain-specific syntheses in neuroscience (Kurth-Nelson et al., 2023), ecology (Holling, 1973), evolutionary biology (Goodenough and Heitman, 2014) and other areas. Previous reviews and perspectives in AI alignment have explored how alignment relates to the diversity of human values, social coordination and human-AI relationships (Dafoe et al., 2020; Gabriel, 2020; Gabriel and Keeling, 2025; Kirk et al., 2024, 2025; Leibo et al., 2026; Sorensen et al., 2024). Technical safety work has reviewed problems in oversight, interpretability and controllability (Anwar et al., 2024; Bereska and Gavves, 2024; Cohen et al., 2024; Ji et al., 2023; Russell, 2019).

Why we are suitable authors for the review

ZKN is an AI researcher and neuroscientist studying representation and reasoning. SS works at the interface of clinical science and cellular biology. JZL and NT contribute expertise in social science and AI safety. EM is a behavioral and cognitive scientist. MGM and MMN are psychiatrists and computational neuroscientists. In each of our respective fields, our work has investigated instances of the mechanisms described in this review. Together we bring interdisciplinary expertise to draw out common principles of healthy living systems across scales and apply them to the problem of AI alignment.

With kind regards,

Zeb Kurth-Nelson
(On behalf of the authors.)